**1** **(a)** Correct elements in all four regions: 2 in the intersection, $4, 6, 8, 10$ in $A$ only, $3, 5, 7$ in $B$ only, and $1, 9$ outside both circles **[B2]**

If 0 scored, allow [B1] for 2 or 3 correct regions

**(b)** $\{2\}$ or 2 **[B1]**

**(c)** 2 **[B1]**

*[Total Marks: 4]*

**2** **(a)** 2, 3, 5, 7 **[B1]**

**(b)** 1, 2, 3, 4, 6, 12 **[B1]**

**(c)** 8 **[B1]**

**(d)** 2, 3 **[B1]**

**(e)** 8 or 9 or 10 **[B1]**

*[Total Marks: 5]*

**3** **(a)** Intersection: $3, 5, 7$ **[B1]**

Remainder of diagram correct: $A$ only region has 2; $B$ only region has $1, 9$; Outer region has $4, 6, 8, 10$ **[B1]**

**(b)** 2 **[B1]**

**(c)** 6 **[B1]**

**(d)** $A \cap B'$ **[B1]**

*[Total Marks: 5]*

**4** **(a)** $1, 2, 3, 6, 9$ **[B1]**

**(b)** $2, 3$ **[B1]**

**(c)** 3 **[B2]**

Partial working: Correctly identifies elements in the union $\{1, 2, 3, 5, 6, 7, 9\}$ OR elements outside $\{4, 8, 10\}$ **[M1]**

**(d)** $\notin$ **[B1]**

*[Total Marks: 5]*

**5** **(a)** 7 **[B3]**

Partial working: Correct simplified equation $7x + 3 = 52$ **[M2]**

Partial working: Sets up unsimplified sum $(2x + 1) + x + (3x - 2) + (x + 4) = 52$ **[M1]**

**(b)** 33 **[B2]**

Partial working: Evaluates correct sub-regions for $x = 7$ (e.g. $15 + 7 + 11$) OR identifies

correct algebraic expression $4x + 5$ **[M1]**

**(c)** Region strictly inside $A$ but outside $B$ is shaded (the left crescent) **[B1]**

*[Total Marks: 6]*

**6 (a)** Intersection: $x$ and Outer region: 4 **[B1]**

$P$ only region: $28 - x$ AND $C$ only region: $22 - x$ **[B1]**

**(b)** $(28 - x) + x + (22 - x) + 4 = 45$ oe **[M1]**

$x = 9$ **[A1]**

**(c)** 19 **[B1]**

*[Total Marks: 5]*

**7 (a)** Summing all regions: $(x + 5) + 2x + x + 3 + 4 + 2 + 5 + (x + 1) = 45$ (or equivalent) **[M1]**

$5x + 20 = 45 \implies 5x = 25$ **[M1]**

$x = 5$ **[A1]**

**(b)** 34 **[B1]**

**(c)** Shading exactly the region representing $P$ only (the top part of circle $P$, excluding the intersections with $Q$ and $R$) **[B1]**

*[Total Marks: 5]*

**8 (a)** 5 in outer region **[B1]**

8 in intersection **[B1]**

17 in F only AND 10 in S only **[B1]**

**(b)** 17 **[B1]**

**(c)** $\frac{27}{40}$ **[B2]**

M1 for $17 + 10$ or 27 seen **[M1]**

**(d)** $\frac{8}{25} \times \frac{7}{24}$ **[M1]**

$\frac{7}{75}$ **[A1]**

*[Total Marks: 7]*

**9 (a)** Intersection of all three: 8 AND Outer region: 3 **[B1]**

Two-set only intersections: 12, 7, and 10 in correct regions **[B1]**

Single set regions: $x$ (in $C$), $2x$ (in $D$), and $x$ (in $B$) **[B1]**

**(b)** $x + 2x + x + 8 + 12 + 7 + 10 + 3 = 100$ oe **[M1]**

$x = 15$ **[A1]**

**(c)** Identifies the regions required: Only $C$ + Only $D$ + ($C \cap D$ only) OR uses $x + 2x + 12$ **[M1]**

57 **[A1]**

*[Total Marks: 7]*

**10 (a)** 4 in center intersection **[B1]**

8, 5, 7 in the two-sport intersections **[B1]**

13, 9, 9 in the one-sport regions AND 5 in outer region **[B1]**

**(b)** 5 **[B1]**

**(c)** 17 **[B1]**

**(d)** $B \cap T' \cap S'$ or $B \cap (T \cup S)'$ **[B2]**

M1 for $B \cap \ldots$ **[M1]**

*[Total Marks: 7]*

**11 (a)** Places 15 in the central intersection **[B1]**

Places 15 in $B \cap T$ only, 13 in $T \cap C$ only, and 7 in $B \cap C$ only **[B1]**

Places $x$, $2x$, $x + 5$, and 8 in the respective remaining regions ($B$ only, $T$ only, $C$ only, and outside the sets) **[B1]**

**(b)** $x + 2x + (x + 5) + 15 + 15 + 13 + 7 + 8 = 103 \implies 4x + 63 = 103$ (or equivalent valid method) **[M1]**

$x = 10$ **[A1]**

**(c)** 85 **[B1]**

**(d)** $\frac{13}{63}$ **[B2]**

If 0 scored, allow [M1] for correctly identifying the denominator $n(T) = 63$ or the numerator $n(T \cap C \cap B') = 13$

*[Total Marks: 8]*

**12 (a)** All 8 regions correctly populated **[B3]**

Partial working: 5 to 7 regions correctly populated **[B2]**

Partial working: 3 to 4 regions correctly populated (e.g. places $12, 15, x$ correctly) **[B1]**

(Correct values are: Center = 12. $M \cap D$ only = 13. $D \cap A$ only = 20. $M \cap A$ only = 16. $M$ only = 9. $D$ only = 20. $A$ only = $x$. Outside = 15)

**(b)** 15 **[B2]**

Partial working: Forms correct sum $9 + 20 + x + 13 + 16 + 20 + 12 + 15 = 120$ OR $105 + x = 120$ **[M1]**

**(c)** 70 **[B2]**

Partial working: Identifies the 20 students in only Drama and Art OR correctly calculates $50 + 20$ **[M1]**

**(d)** Region strictly inside $M$ but outside both $D$ and $A$ is shaded (the top crescent / "only $M$" region) **[B1]**

*[Total Marks: 8]*